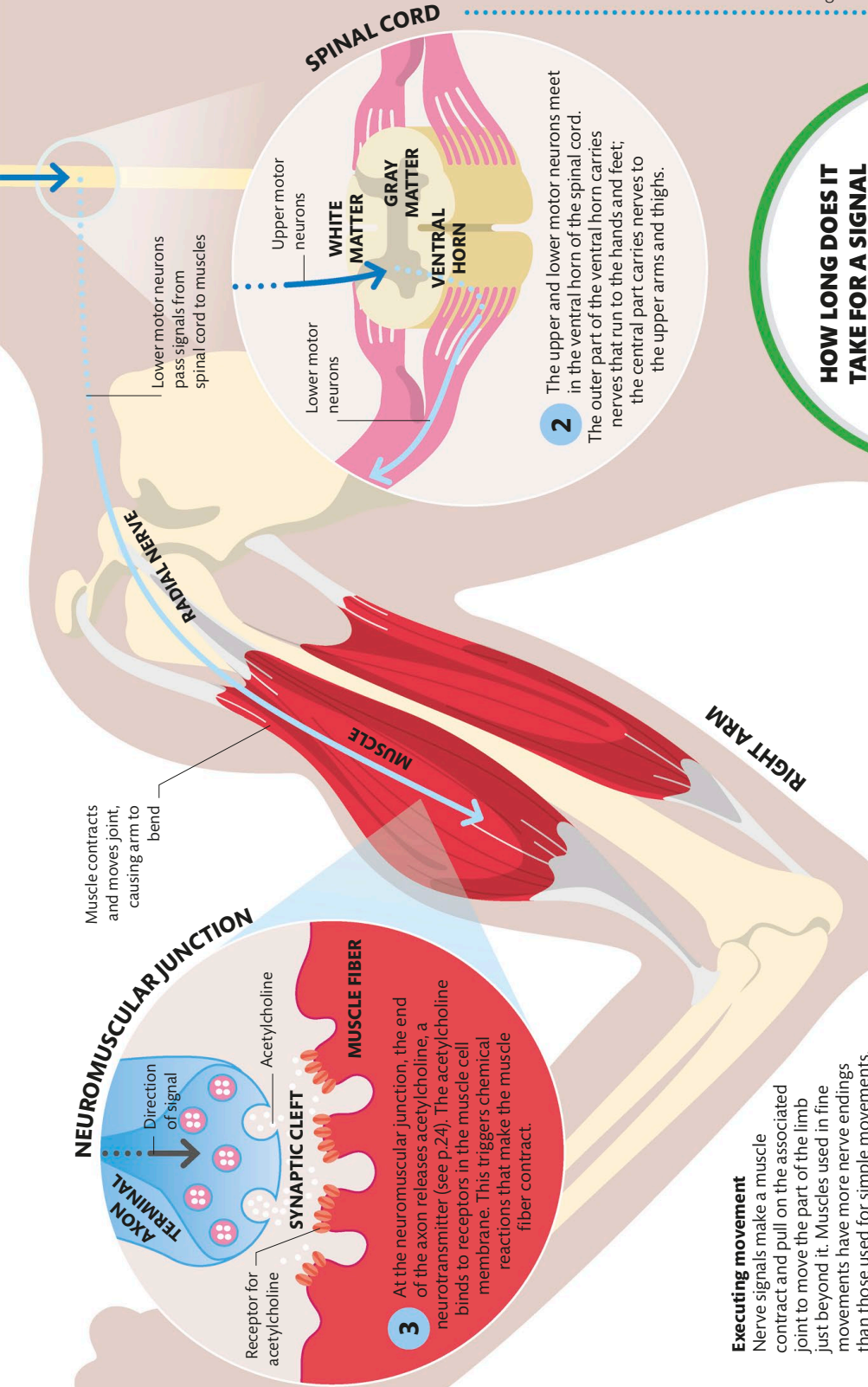




2 The upper and lower motor neurons meet in the ventral horn of the spinal cord. The outer part of the ventral horn carries nerves that run to the hands and feet; the central part carries nerves to the upper arms and thighs.

HOW LONG DOES IT TAKE FOR A SIGNAL TO TRAVEL FROM BRAIN TO MUSCLE?

Signals can travel from the brain to our muscles at a speed of up to 395 ft (120 m) per second.

Executing movement

Nerve signals make a muscle contract and pull on the associated joint to move the part of the limb just beyond it. Muscles used in fine movements have more nerve endings than those used for simple movements.

From spine to muscle

Inside the spinal cord, the axons of the corticospinal tract, which are covered with a myelin sheath, form the white matter. The gray matter at the center of the spinal cord consists of the cell bodies of motor neurons. The ends of the corticospinal axons (known as upper motor neurons) synapse on to motor neurons (known as lower motor neurons) in the ventral horn of the gray matter. The axons of the lower neurons exit the spine through gaps in the vertebrae (see p.12) and extend to the muscle fibers. The point where the nerve endings activate the muscle fibers to complete the movement is called the neuromuscular junction.

3 At the neuromuscular junction, the end of the axon releases acetylcholine, a neurotransmitter (see p.24). The acetylcholine binds to receptors in the muscle cell membrane. This triggers chemical reactions that make the muscle fiber contract.